

Chapter 5 - System Implementation

5.1 Introduction

The implementation phase involved developing the AI-Powered Customer Support & Service Automation application using the Salesforce Lightning Platform. The application was implemented by creating custom objects, configuring fields and relationships, applying validation rules, developing automation using Salesforce Flows, configuring queues for case assignment, and integrating Artificial Intelligence through Salesforce Prompt Builder.

The implementation followed a modular approach, ensuring that each component was independently configured and thoroughly tested before integrating it into the overall system.

5.2 Salesforce Environment Setup

The development environment was established using Salesforce Developer Edition, providing access to Salesforce Lightning Experience and AI capabilities. The Salesforce platform was configured to support custom object development, automation, AI integration, and security management.

The following Salesforce tools were used:

Tool	Purpose
Salesforce Developer Edition	Development Platform
Object Manager	Custom Object Configuration
Lightning App Builder	User Interface Development
Flow Builder	Business Process Automation
Prompt Builder	AI Summary Generation
Queues	Automatic Case Assignment
Validation Rules	Data Validation
Reports & Dashboards	Data Analysis

5.3 Custom Object Implementation

Four custom objects were created to represent the core entities of the application.

Customer

The Customer object stores customer information including contact details and customer type.

Fields implemented:

- Customer Name
- Email
- Phone
- Customer Type

The screenshot shows the Salesforce Setup interface for the 'Customer' object. The 'Fields & Relationships' tab is selected, displaying a list of 7 fields. The fields are: Created By, Customer Name, Customer Type, Email, Last Modified By, Owner, and Phone. Each field has a corresponding field name, data type, and a checkbox for indexing. The 'Customer Name' field is highlighted in blue.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Customer Name	Name	Text(80)		✓
Customer Type	Customer_Type__c	Picklist		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		

Product

The Product object stores product information including warranty and category details.

Fields implemented:

- Product Name
- Product Code
- Category
- Warranty (Months)
- Active

Developer Edition

Search Setup

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Merchandise

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Fields & Relationships

9 Items, Sorted by Field Label

Q Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Active	Active__c	Checkbox		
Category	Category__c	Picklist		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Merchandise Name	Name	Text(80)		✓
Owner	OwnerId	Lookup(User,Group)		✓
Product Code	Product_Code__c	Text(50)		
Product Name	Product_Name__c	Text(225)		
Warranty (Months)	Warranty_Months__c	Number(3, 0)		

Order

The Order object stores customer purchase information and establishes relationships between customers and products.

Fields implemented:

- Order Number
- Order Date
- Status
- Customer Lookup
- Product Lookup

Developer Edition

Search Setup

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Order

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Q Quick Find

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FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Customer	Customer__c	Lookup(Customer)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Order Date	Order_Date__c	Date		
Order Name	Name	Text(80)		✓
Order Number	Order_Number__c	Auto Number		
Owner	OwnerId	Lookup(User,Group)		✓
Product	Product__c	Lookup(Merchandise)		✓
Status	Status__c	Picklist		

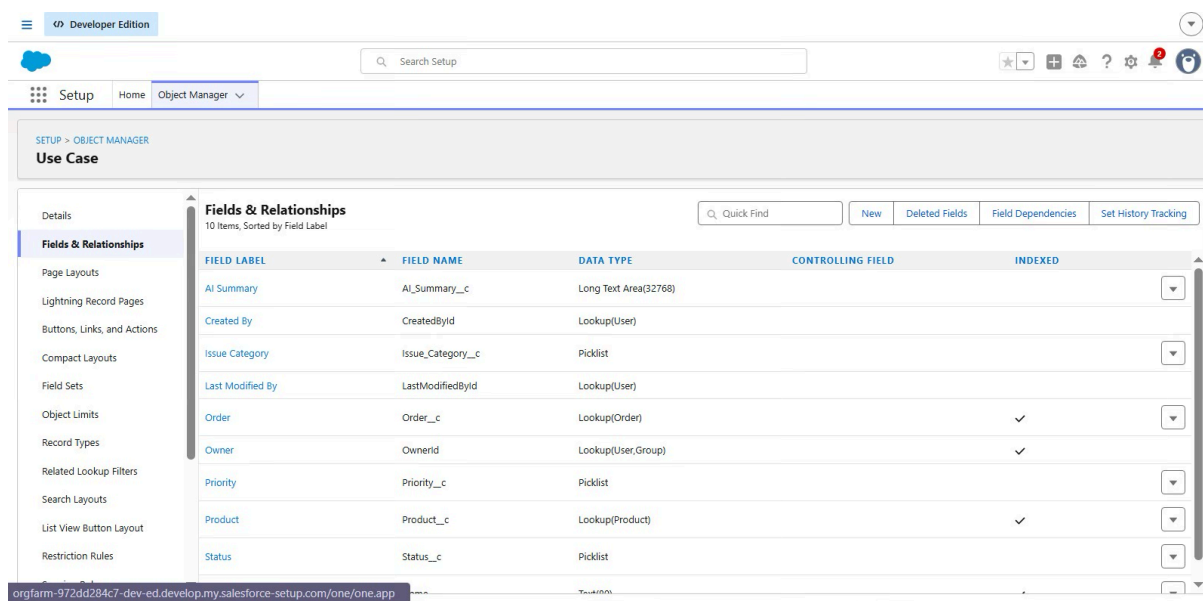
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Use Case

The Use Case object stores customer support requests and AI-generated summaries.

Fields implemented:

- Product Lookup
- Order Lookup
- Priority
- Issue Category
- AI Summary



FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
AI Summary	AI_Summary__c	Long Text Area(32768)		
Created By	CreatedById	Lookup(User)		
Issue Category	Issue_Category__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Order	Order__c	Lookup(Order)		✓
Owner	OwnerId	Lookup(User,Group)		✓
Priority	Priority__c	Picklist		
Product	Product__c	Lookup(Product)		✓
Status	Status__c	Picklist		

5.4 Field Configuration

Custom fields were configured using Salesforce Object Manager. Appropriate data types such as Text, Email, Phone, Number, Checkbox, Picklist, Lookup Relationship, Date, Auto Number, and Long Text Area were selected according to business requirements.

Lookup relationships were established between Customer, Product, Order, and Use Case objects to maintain data consistency and support relational data management.

5.5 Validation Rule Implementation

Validation Rules were implemented to ensure accurate and consistent data entry.

Damage_Requires_Priority

This validation rule prevents users from creating Damage cases without selecting a Priority.

Business Rule: Damage cases must always have a priority assigned.

Warranty_Delivery_Requires_Priority

This validation rule ensures that Warranty and Delivery cases also require a Priority before the record can be saved.

Business Rule: Warranty and Delivery cases cannot be created without specifying Priority.

The screenshot shows the Salesforce Setup interface in Developer Edition. The navigation menu on the left includes Setup, Home, and Object Manager. The main content area is titled 'Use Case' and displays the configuration for a 'Use Case Validation Rule' named 'Damage_Requires_Priority'. The rule is active and includes an error condition formula, error message, and description. The 'Created By' and 'Modified By' fields show the user 'VELLAMPALLI VENKATANIHKIL'.

Validation Rule Detail			
Rule Name	Damage_Requires_Priority	Active	✓
Error Condition Formula	AND(ISPICKVAL(Issue_Category__c,"Damage"), ISBLANK(TEXT(Priority__c)))		
Error Message	Priority is mandatory for Damage cases.		
Description	Ensures Priority is mandatory for Damage cases.		
Created By	VELLAMPALLI VENKATANIHKIL AP24110010587, 7/6/2026, 10:09 AM	Modified By	VELLAMPALLI VENKATANIHKIL AP24110010587, 7/6/2026, 10:09 AM

5.6 Flow Implementation

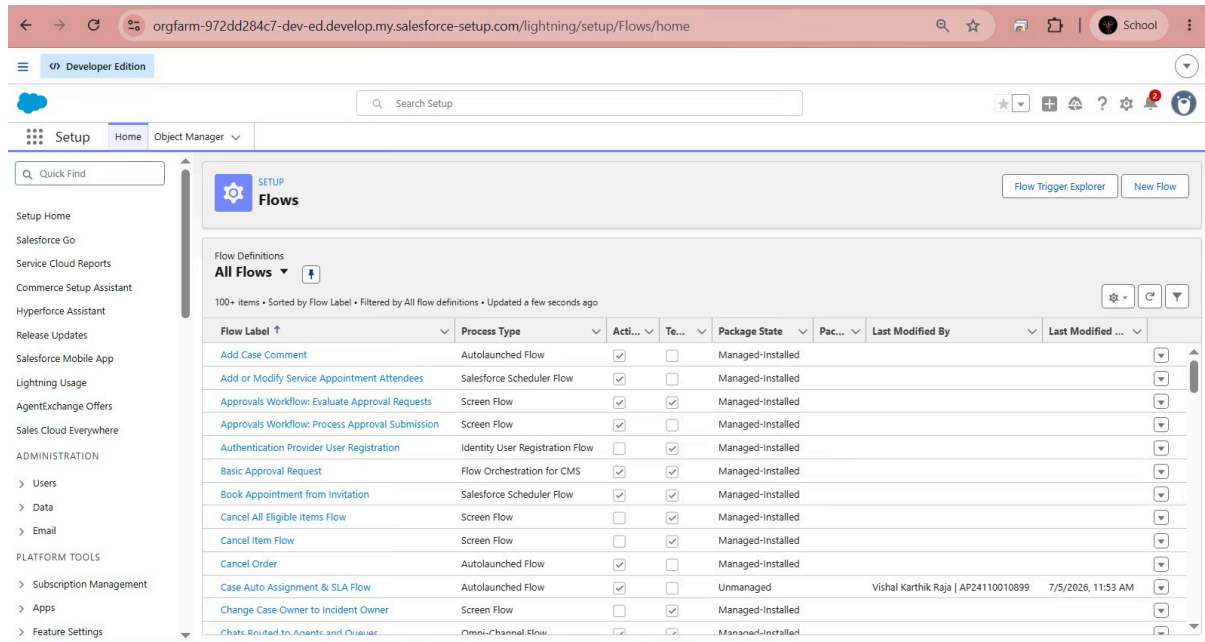
Salesforce Record-Triggered Flows were implemented to automate customer support operations.

Case Assignment Flow

This flow executes automatically whenever a Use Case record is created or updated. It evaluates the Issue Category and performs decision-based processing according to predefined business logic.

Case Auto Assignment & SLA Flow

This flow executes when a new Use Case is created. If the Priority is marked as High, the flow automatically assigns ownership of the case to the Damage Support Queue, reducing manual assignment efforts and improving response time.



The screenshot shows the Salesforce Setup interface with the 'Flows' section selected. The 'Case Auto Assignment & SLA Flow' is highlighted in the list of flow definitions. The table below represents the data shown in the screenshot.

Flow Label	Process Type	Activated	Tested	Package State	Package	Last Modified By	Last Modified
Add Case Comment	Autolaunched Flow	✓	✓	Managed-Installed			
Add or Modify Service Appointment Attendees	Salesforce Scheduler Flow	✓	✓	Managed-Installed			
Approvals Workflow: Evaluate Approval Requests	Screen Flow	✓	✓	Managed-Installed			
Approvals Workflow: Process Approval Submission	Screen Flow	✓	✓	Managed-Installed			
Authentication Provider User Registration	Identity User Registration Flow	✓	✓	Managed-Installed			
Basic Approval Request	Flow Orchestration for CMS	✓	✓	Managed-Installed			
Book Appointment from Invitation	Salesforce Scheduler Flow	✓	✓	Managed-Installed			
Cancel All Eligible Items Flow	Screen Flow	✓	✓	Managed-Installed			
Cancel Item Flow	Screen Flow	✓	✓	Managed-Installed			
Cancel Order	Autolaunched Flow	✓	✓	Managed-Installed			
Case Auto Assignment & SLA Flow	Autolaunched Flow	✓	✓	Unmanaged		Vishal Karthik Raja AP24110010899	7/5/2026, 11:53 AM
Change Case Owner to Incident Owner	Screen Flow	✓	✓	Managed-Installed			
Chatbot Routing to Agents and Queues	Omni-Channel Flow	✓	✓	Managed-Installed			

5.7 Queue Implementation

Salesforce Queues were configured to distribute customer support requests among support teams.

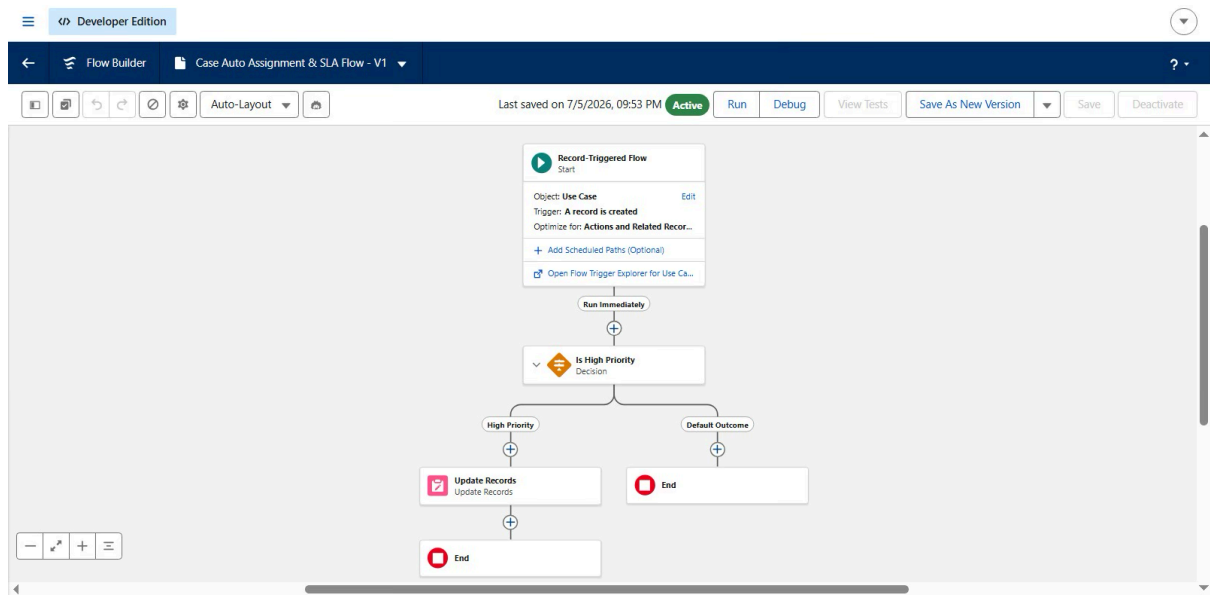
Damage Support Queue

Handles high-priority customer support requests related to damaged products.

Warranty Support Queue

Handles customer requests related to warranty claims and warranty services.

Queue implementation improves workload distribution and reduces manual case assignment.



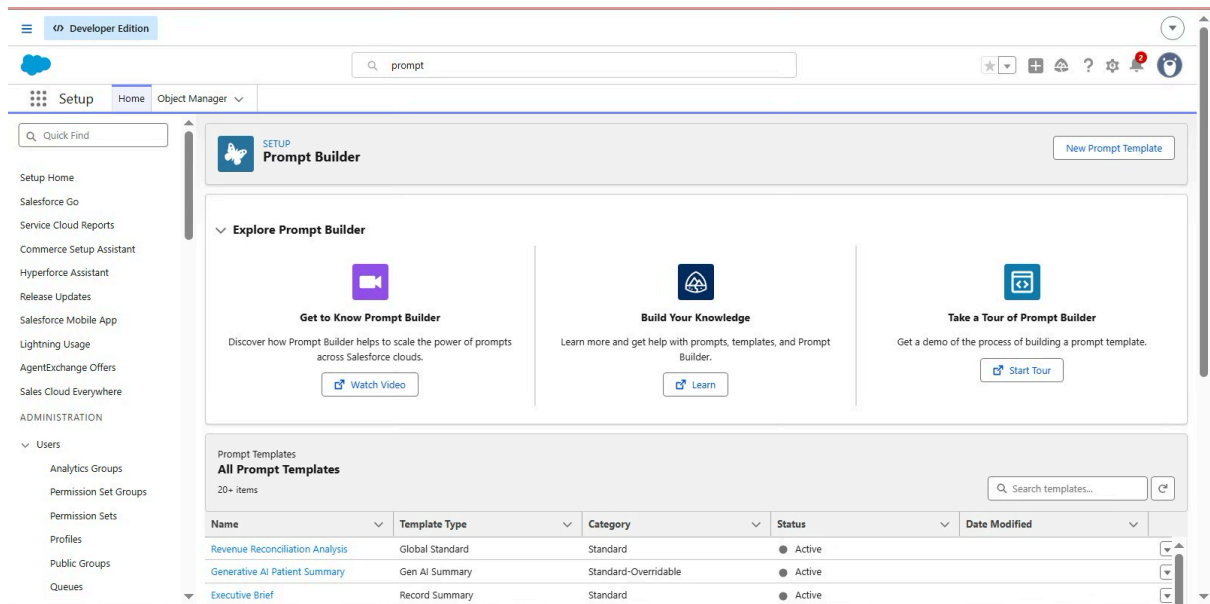
5.8 Prompt Builder Implementation

Salesforce Prompt Builder was configured to generate AI-powered customer support summaries.

Configuration Details:

Parameter	Value
Prompt Template	MetaDataInspection
Prompt Type	Field Generation
Object	Use Case
Output Field	AI Summary

The prompt template generates concise and professional summaries based on the available customer support information, improving documentation quality and assisting support personnel.



5.9 Lightning Application Implementation

A custom Lightning Application named AI-Powered Customer Support & Service Automation was developed using Salesforce Lightning App Builder.

The application provides centralized access to:

- Customer
- Product
- Order
- Use Case
- Reports
- Dashboards

The user interface was designed to simplify navigation while providing quick access to frequently used business objects.

5.10 Security Implementation

Security mechanisms were configured using Salesforce security features.

Implemented security controls include:

- Profiles
- Queue Ownership
- Field-Level Security
- Validation Rules
- Lightning Page Permissions

These controls ensure that only authorized users can access or modify application data.

5.11 AI Integration

Artificial Intelligence was integrated into the application using Salesforce Prompt Builder.

The AI component automatically generates professional summaries for customer support cases, reducing manual documentation efforts and ensuring consistent communication across support teams.

The AI Summary field stores generated content directly within the Use Case record, making it available for future reference.

5.12 Implementation Summary

The implementation phase successfully transformed the project requirements into a fully functional Salesforce application. Through the use of custom objects, relationships, validation rules, automation flows, queues, Lightning components, and AI-powered Prompt Builder, the system automates customer support processes while improving efficiency, data accuracy, and user productivity.

The completed application demonstrates the practical implementation of Salesforce CRM capabilities combined with Artificial Intelligence to build a scalable customer support solution.